

Verification Data Record

Date: Dec-15-2020 18:18:51

Customer Name: Ferretti Validaciones Work Order: 18044-2012093
Service Notes: ID: Ferretti L3

Unit Under Test

Instrument: LasairIII_310C Serial Number: 61837
Firmware Version: PCE: B2B53624 1.9 Laser Serial Number: No Change

Environmental

Temperature: 24 Deg C Relative Humidity: 46%

Instrument Condition Received

☒ In-Tolerance ☐ Out-of-Tolerance ☐ Operational Failure

Threshold Information

<u>Channel Size (micron)</u>	<u>Measured Threshold/Gain</u>	<u>Received Threshold/Gain</u>	<u>Measurement Type</u>
Ch 1: 0.30	00702 / 0	00722 / 0	Distribution
Ch 2: 0.50	02698 / 0	02763 / 0	Distribution
Ch 3: 1.00	00705 / 1	00714 / 1	Distribution
Ch 4: 5.00	05610 / 1	05815 / 1	Distribution
Ch 5: 10.00	01331 / 2	01338 / 2	Extrapolate
Ch 6: 25.00	06545 / 2	06584 / 2	Extrapolate

Verification Results

<u>Parameter</u>	<u>Results</u>	<u>Allowed Range</u>	<u>Expanded Uncertainty</u>	<u>Pass/Fail/ Skipped</u>
Flow Rate	28.4 L/min	26.9 to 29.7 L/min	n/a	Pass
Pre Zero Count	0.00 Cnt/m ³	0 to 176.50 Cnt/m ³	n/a	Pass (*1)
Size Error @ 0.30	0.8%	-10.0% to 10.0%	2.06%	Pass (*2)
Size Error @ 0.50	1.3%	-10.0% to 10.0%	2.04%	Pass (*2)
Size Error @ 1.00	0.9%	-10.0% to 10.0%	2.20%	Pass (*2)
Size Error @ 5.00	2.0%	-10.0% to 10.0%	1.55%	Pass (*2)
Size Error @ 10.00	0.3%	-10.0% to 10.0%	1.95%	Pass (*2)
Size Error @ 25.00	0.3%	-10.0% to 10.0%	1.95%	Pass (*2)
Resolution @ 0.30	8.8%	0.0% to 15.0%	n/a	Pass
CE(%) @ 0.30	51.0%	30.0% to 70.0%	n/a	Pass
CE(%) @ 0.50	91.1%	90.0% to 110.0%	n/a	Pass

Note *1: Process staged intervals to achieve 95% UCL for stated concentration

Note *2: Sizing Error operational sign per ISO 21501-4

Performed by

Technician Name: Favio Wainstein

FAVIO J. WAINSTEIN
Director Técnico
Dirección de Metrología
HiTEC S.R.L.

Certificate of Calibration

Date: Dec-15-2020 18:55:06

Customer Name: Ferretti Validaciones Work Order: 18044-2012093
Service Notes: ID: Ferretti L3

Unit Under Test

Instrument: LasairIII_310C Serial Number: 61837
Firmware Version: PCE: B2B53624 1.9 Calibration Due: Dec-15-2021

Environmental

Temperature: 24 Deg C Relative Humidity: 44%

Certification

Particle Measuring Systems certifies that the instrument listed above meets or exceeds manufacturing specifications and meets the requirements of ISO 21501-4:2007. It has been calibrated using equipment and/or standards whose accuracies are traceable to the National Institute of Standards and Technology (NIST), or have been derived from acceptable values of natural physical constants, or by the ratio type of self-calibration. This instrument was calibrated in accordance with the Particle Measuring Systems quality system documentation which meets ISO9001:2008 requirements. This certificate may not be reproduced, except in full, without written consent from Particle Measuring Systems.

Calibration Standards

Procedure Used: 2793 Rev E Program Used: 1000013097 Rev K

Instrument Condition Received

☒ In-Tolerance ☐ Out-of-Tolerance ☐ Operational Failure

Particles Used

<u>Particle Size</u>	<u>Std Deviation</u>	<u>Lot Number</u>	<u>Expiration Date</u>	<u>Uncertainty</u>
0.303 micron	0.005	204665	Nov-30-2021	0.006
0.496 micron	0.009	203144	Oct-31-2021	0.008
0.994 micron	0.010	192848	Jan-31-2021	0.015
5.022 micron	0.050	205967	Dec-31-2021	0.039

References Used

<u>Type</u>	<u>Serial Number</u>	<u>Calibration Due</u>
LasairII-110	114618	Feb-28-2021
Flow Meter	40401638044	Aug-31-2021

Performed by

Technician Name: Favio Wainstein

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Calibration Data Record

Date: Dec-15-2020 18:55:06

Customer Name: Ferretti Validaciones Work Order: 18044-2012093
Service Notes: ID: Ferretti L3

Unit Under Test

Instrument: LasairIII_310C Serial Number: 61837
Firmware Version: PCE: B2B53624 1.9 Laser Serial Number: No Change
Calibration Due: Dec-15-2021

Environmental

Temperature: 24 Deg C Relative Humidity: 44%

Threshold Information

Channel Size (micron)	Measured Threshold/Gain	Measurement Type
Ch 1: 0.30	00714 / 0	Distribution
Ch 2: 0.50	02710 / 0	Distribution
Ch 3: 1.00	00703 / 1	Distribution
Ch 4: 5.00	05671 / 1	Distribution
Ch 5: 10.00	01358 / 2	Extrapolate
Ch 6: 25.00	06693 / 2	Extrapolate

Calibration Results

Parameter	Results	Allowed Range/Condition	
Flow Rate	28.4 L/min	27.5 to 29.2 L/min	
Resolution @ 0.30	10.5%	0.0% to 15.0%	
CE(%) @ 0.30	54.4%	30.0% to 70.0%	
CE(%) @ 0.50	92.4%	90.0% to 110.0%	
Zero Count	0.00 Cnt/m ³	0 to 7.07 Cnt/m ³	(*1)
Sampling Time	0.84%	0.0% to 1.0%	(*2)
Response Rate	0.003%	0.0 % to 0.5%	(*2)
Maximum particle number concentration	48743777 Cnt/m ³	10% coincidence loss	(*2)

Parameter	Results	Size Error	Expanded Uncertainty	
Ch 1 Size	0.30 micron	-0.27%	2.05%	(*3)
Ch 2 Size	0.50 micron	-0.62%	2.04%	(*3)
Ch 3 Size	1.00 micron	-0.80%	2.20%	(*3)
Ch 4 Size	5.00 micron	-0.66%	1.54%	(*3)
Ch 5 Size	10.00 micron	-0.87%	1.91%	(*3)
Ch 6 Size	25.00 micron	-0.87%	1.91%	(*3)

Notes

Note *1: Process staged intervals to achieve 95% UCL for stated concentration
Note *2: Tested and verified during design and development
Note *3: Sizing Error operational sign per ISO 21501-4

CERTIFICATE OF SOFTWARE VALIDATION

Assurance of Product Lifecycle Integrity

Particle Measuring Systems assures that the software has been designed, developed, tested, and validated according to the "Quality and Project Plan" for this product.

The software cycle milestones are reviewed and approved by management.

This product meets the requirements specification and passed all prerequisite tests prior to release. Details of product testing are defined in the supplied software release note.

To support this certificate for GAMP, cGMP, and ISO 9001 requirements of the user of this product, Particle Measuring Systems, Inc. will make the following documents available to customers.

- Requirements Document
- Project and Quality Plan
- Software Design Document
- Test Plans and Results
- Software Release Notes
- Software Codes of Practice

Particle Measuring Systems, Inc. will maintain the software lifecycle documents and issue those documents identified above to the customer for their own use, as verification of completion of testing for the version applicable to the licensed copy.

Pharm Net RTS
Product Name:

3.0.0
Revision Number:



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6/22/04
Date

Mark Hallworth
Mark Hallworth, Software Product Manager

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